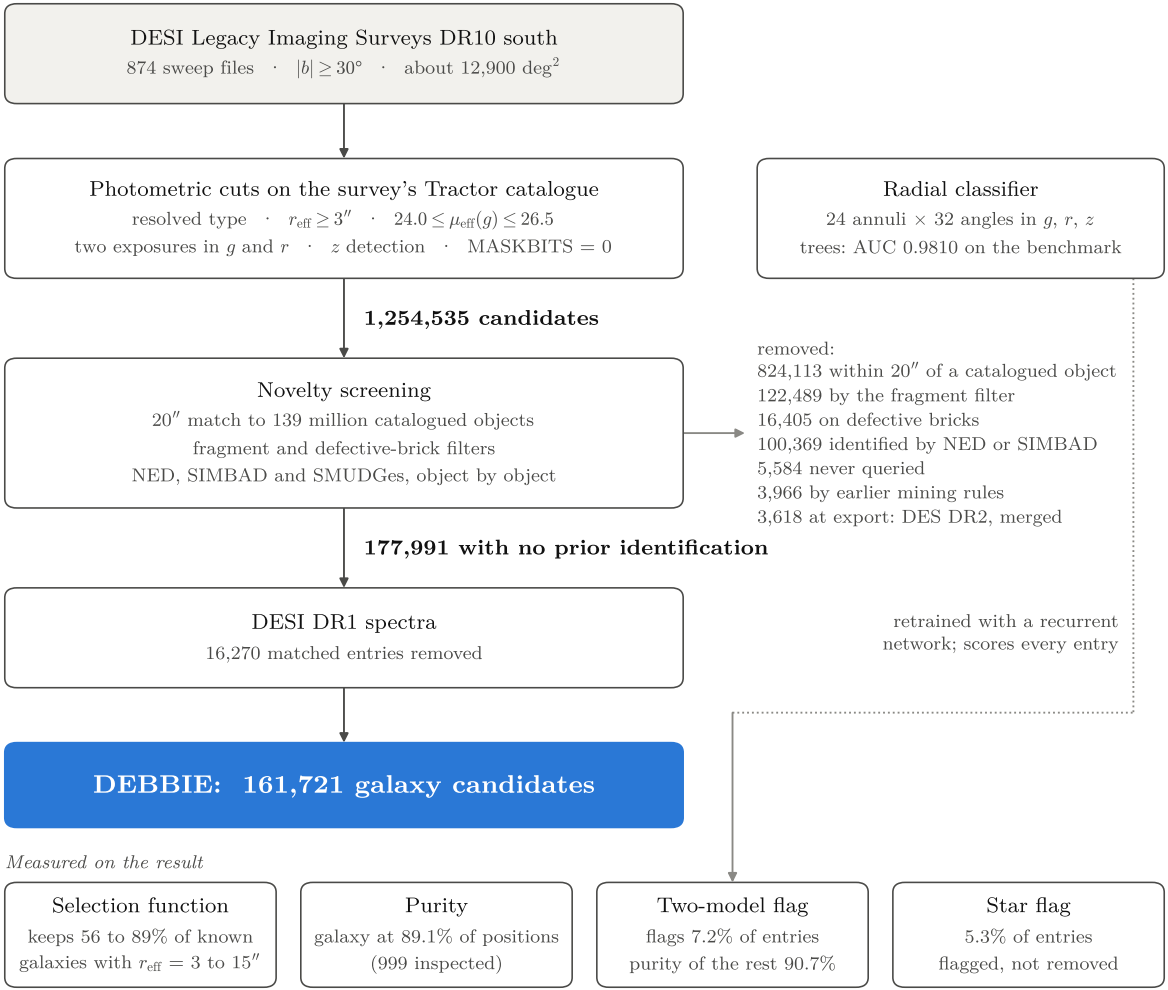

DESI Legacy Imaging Surveys DR10 south
874 sweep files · $|b| \geq 30°$ · about 12,900 deg^2
Photometric cuts on the survey's Tractor catalogue
resolved type · $r_{\rm eff} \geq 3''$ · $24.0 \leq \mu_{\rm eff}(g) \leq 26.5$
two exposures in g and r · z detection · MASKBITS = 0
Radial classifier
24 annuli × 32 angles in g, r, z
trees: AUC 0.9810 on the benchmark
1,254,535 candidates
Novelty screening
20″ match to 139 million catalogued objects
fragment and defective-brick filters
NED, SIMBAD and SMUDGes, object by object
removed:
824,113 within 20″ of a catalogued object
122,489 by the fragment filter
16,405 on defective bricks
100,369 identified by NED or SIMBAD
5,584 never queried
3,966 by earlier mining rules
3,618 at export: DES DR2, merged
177,991 with no prior identification
DESI DR1 spectra
16,270 matched entries removed
retrained with a recurrent
network; scores every entry
DEBBIE: 161,721 galaxy candidates
Measured on the result
Selection function
keeps 56 to 89% of known
galaxies with $r_{\rm eff}$ = 3 to 15″
Purity
galaxy at 89.1% of positions
(999 inspected)
Two-model flag
flags 7.2% of entries
purity of the rest 90.7%
Star flag
5.3% of entries
flagged, not removed